Rockchip NPU Model Benchmark

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Rockchip Electronics Co., Ltd.

No.18 Building, A District, No.89, Software Boulevard Fuzhou, Fujian, PRC

Website: www.rock-chips.com

Customer service Tel: +86-4007-700-590

Customer service Fax: +86-591-83951833

Customer service e-Mail: fae@rock-chips.com

Preface

Product Versions

Chip Models	SDK Version
RK1820, RK1828, RK3572, RK3588, RK3576, RV1126B, RK3562, RV1109, RK3566, RV1126, RK1808, RK3568	General

Intended Audience

This document is mainly intended for:

Technical support engineers

Software development engineers

Revision History

Date	Version	Author	Description
2026-06-05	V1.0.0	Cherry Chen	Initial version

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1. Overview

This document summarizes model benchmark across various Rockchip NPU chips, covering CNN/ViT, LLM, and VLM model types. The data is based on internal testing and is provided for reference only.

2. Chip Overview

Chip	NPU Cores	Architecture
RK1820/RK1828	8 cores	RKNN3
RK3588	3 cores	RKNN2
RK3576	2 cores	RKNN2
RK3572	1 core	RKNN3
RK3566	1 core	RKNN2
RK3568	1 core	RKNN2
RK3562	1 core	RKNN2
RV1126B	1 core	RKNN2
RV1126	1 core	RKNN2
RV1109	1 core	RKNN2
RK1808	1 core	RKNN2

3. Model Overview

Category	Model List
Object Detection	ppyoloe(m/s), retinaface-mobile320, retinaface-resnet50-320, yolo11(m/n/s), yolo_world_v2s, yolo_world_v2s-clip-text, yolov10(n/s), yolov5(m/n/s), yolov5s_relu, yolov6(m/n/s), yolov7, yolov7-tiny, yolov8(m/n/s), yolov8n-obb, yolov8n-pose, yolov9s, yolox(m/s)
Classification/Recognition	efficientnet_v2_m, iresnet50, lite-transformer-decoder-16, lite-transformer-encoder-16, lprnet, mobilenetv1, mobilenetv2, pfld, ppocrv4_det, ppocrv4_rec, resnet34, resnet50-v2-7
Image Segmentation	deeplab-v3-plus-mobilenet-v2, mobilesam(decoder/encoder/encoder_tiny), pp_liteseg_cityscapes, ppseg_lite, yolov5-seg(m/n/s), yolov8-seg(m/n/s)

Category	Model List
ViT/CLIP	clip-vit-base-patch32-text, clip-vit-base-patch32-vision, cn-clip-large-patch14(text/vision), dinov2_vit14(b/s), dinov3_vit16(b/s), dinov3_vit16-640p, EVA02-Large, fastvit_sa12, siglip, siglip2(patch14-384/patch16-512), vit(patch14)(base/large), vit(patch16)(base/large)
LLM	Llama-3.2-3B-Instruct, Qwen2.5(0.5B/0.5B-Instruct/1.5B-Instruct/3B/3B-Instruct/7B/7B-Instruct), Qwen3(0.6B/1.7B/4B/4B-Instruct-2507/8B), Qwen3.5-2B
VLM	FastVLM-1.5B, InternVL3-2B, InternVL3.5-4B, MiniCPM-V-4, Qwen2.5-Omni-3B, Qwen2.5-VL-3B, Qwen2.5-VL-7B_prune, Qwen3-VL-2B, Qwen3-VL-2B-Instruct, Qwen3-VL-4B

> A total of 96 models, 651 performance records

4. CNN/ViT Model Performance

Below are the benchmarks for CNN and ViT models on the Rockchip NPU, organized into four sub-categories: detection, classification, segmentation, and ViT/CLIP.

> Metrics: Single Core FPS = single NPU core inference frame rate, Single-Core Latency = per-frame inference time, Multi-Core FPS = multi-core parallel inference frame rate

4.1 Object Detection Model Performance

Below are the benchmarks for object detection models on Rockchip NPU.

RK1820/RK1828

Model Name	Input	Quantization Method	Single Core FPS	Single Core Latency(ms)	Multi Core FPS	Memory(MB)
yolo11n	640x640	W8A8	42.200	23.700	118.690	9.690
yolov5m	640x640	W8A8	18.460	54.170	63.130	32.128
yolov5n	384x288	W8A8	120.870	8.273	262.200	3.875
yolov5n	416x256	W8A8	128.830	7.762	295.270	3.801
yolov5n	640x640	W8A8	42.870	23.329	92.320	9.293
yolov8n	640x640	W8A8	42.250	23.670	100.840	10.176
yolov8n-pose	640x640	W8A8	44.770	22.337	142.020	10.590
yolov6n	640x640	W8A8	52.830	18.928	138.510	8.814
yolov5s	480x288	W8A8	81.540	12.264	186.930	10.004
yolov9s	640x640	W8A8	17.850	56.021	108.110	13.518
yolov5s	640x640	W8A8	35.630	28.068	220.600	58.098
yolox_s	288x384	W8A8	96.560	10.356	452.280	10.729
yolo11s	640x640	W8A8	29.690	33.684	109.400	17.953
yolov8s	640x640	W8A8	29.040	34.430	105.090	17.966
yolox_s	640x640	W8A8	32.880	30.416	142.480	16.146
yolov6s	640x640	W8A8	30.720	32.555	202.670	35.891
yolov8m	640x640	W8A8	15.110	66.169	-	44.448

RK3572

Model Name	Input	Quantization Method	Single Core FPS	Single Core Latency(ms)	Memory(MB)
yolo11s	640x640	W8A8	41.920	23.857	17.953
yolov5m	640x640	W8A8	24.480	40.851	32.128
yolov5s	640x640	W8A8	56.870	17.584	14.416
yolov6n	640x640	W8A8	103.170	9.693	8.814
yolov6s	640x640	W8A8	36.740	27.220	23.640
yolov8m	640x640	W8A8	17.320	57.725	44.448
yolov8s	640x640	W8A8	42.240	23.672	17.966
yolox_s	640x640	W8A8	41.950	23.839	16.146
yolov9s	640x640	W8A8	39.720	25.180	-
yolo11n	640x640	W8A8	71.590	13.970	9.684
yolo11m	640x640	W8A8	16.830	59.410	36.754
yolov5s_relu	640x640	W8A8	66.640	15.010	14.416
yolov6m	640x640	W8A8	18.590	53.790	43.203
yolov7-tiny	640x640	W8A8	77.570	12.890	12.756
yolov7	640x640	W8A8	14.360	69.620	55.297
yolov8n-obb	640x640	W8A8	82.250	12.160	10.080
yolov10n	640x640	W8A8	72.720	13.750	9.390
yolov10s	640x640	W8A8	42.830	23.350	14.367
retinaface-mobile320	320x320	W8A8	462.110	2.160	2.426
ppyoloe_s	640x640	W8A8	25.550	39.140	16.362
yolov8n	640x640	W8A8	75.030	13.330	10.174
yolov5n	640x640	W8A8	92.780	10.780	9.293
yolo_world_v2s-clip-text	1x20	W8A8	147.610	6.770	121.240

RK3588

Model Name	Input	Quantization Method	Single Core FPS	Single Core Latency(ms)	Multi Core FPS	Memory(MB)
ppyoloe_s	640x640	W8A8	38.660	25.870	108.430	41.200
retinaface-mobile320	320x320	W8A8	301.210	3.320	602.650	41.790
yolo11n	640x640	W8A8	62.070	16.110	167.410	19.280
yolo11s	640x640	W8A8	34.490	28.990	92.170	44.280
yolov10n	640x640	W8A8	62.970	15.880	171.930	18.100
yolov10s	640x640	W8A8	34.410	29.060	95.140	35.880
yolov5m	640x640	W8A8	21.300	46.950	59.450	85.550
yolov5n	640x640	W8A8	85.860	11.650	234.570	22.360
yolov5s_relu	640x640	W8A8	98.940	10.110	185.940	37.850
yolov6m	640x640	W8A8	28.260	35.380	50.790	124.490
yolov6n	640x640	W8A8	164.910	6.060	300.940	23.800
yolov7-tiny	640x640	W8A8	104.450	9.570	203.370	32.540
yolov8m	640x640	W8A8	16.580	60.310	45.980	98.110
yolov8n	640x640	W8A8	75.010	13.330	205.890	19.950
yolov8n-obb	640x640	W8A8	76.060	13.150	208.580	20.070
yolox_m	640x640	W8A8	16.000	62.500	-	-
ppyoloe_m	640x640	W8A8	15.800	63.290	-	-
retinaface-resnet50-320	320x320	W8A8	49.200	20.330	-	-
yolov5s	640x640	W8A8	47.860	20.890	135.130	37.940
yolov6s	640x640	W8A8	36.090	27.710	104.370	67.370
yolov7	640x640	W8A8	11.370	87.920	32.800	149.840
yolo11m	640x640	W8A8	12.780	78.240	35.560	89.670
yolox_s	640x384	W8A8	36.880	27.120	105.140	43.680
yolov9s	640x640	W8A8	29.040	34.430	83.090	39.130
yolov8n-pose	640x640	W8A8	64.048	15.610	170.700	22.900
yolov8s	640x640	W8A8	38.581	25.920	107.010	46.270

RK3576

Model Name	Input	Quantization Method	Single Core FPS	Single Core Latency(ms)	Multi Core FPS	Memory(MB)
yolov8s	640x640	W8A8	40.440	24.730	76.430	33.940
yolov5s	640x640	W8A8	56.000	17.860	107.380	27.820
yolov6s	640x640	W8A8	33.080	30.230	65.230	44.070
yolov8n-pose	640x640	W8A8	75.820	13.190	140.700	18.350
yolox_s	640x384	W8A8	32.670	30.610	55.110	35.610
yolox_m	640x640	W8A8	17.600	56.820	-	-
ppyoloe_m	640x640	W8A8	17.800	56.180	-	-
retinaface-resnet50-320	320x320	W8A8	56.600	17.670	-	-
yolov5n	640x640	W8A8	119.540	8.370	177.080	17.800
yolov5s_relu	640x640	W8A8	66.850	14.960	103.780	23.200
yolov5m	640x640	W8A8	23.680	42.230	39.800	60.100
yolov6n	640x640	W8A8	108.580	9.210	174.310	15.600
yolov6m	640x640	W8A8	16.880	59.260	28.360	83.500
yolov7	640x640	W8A8	12.500	80.000	20.620	101.800
yolov7-tiny	640x640	W8A8	78.100	12.800	112.440	19.900
yolov8n	640x640	W8A8	93.500	10.700	133.100	16.600
yolov8m	640x640	W8A8	15.930	62.780	27.000	73.700
yolov8n-obb	640x640	W8A8	95.450	10.480	123.680	16.900
yolox_s	640x640	W8A8	42.240	23.680	67.370	33.200
yolov10n	640x640	W8A8	84.250	11.870	116.600	17.700
yolov10s	640x640	W8A8	40.010	25.000	58.990	29.400
yolo11n	640x640	W8A8	82.860	12.070	109.750	19.800
yolo11s	640x640	W8A8	39.060	25.600	58.770	34.600
yolo11m	640x640	W8A8	14.400	69.450	25.720	65.500
ppyoloe_s	640x640	W8A8	47.210	21.180	70.010	31.300
retinaface-mobile320	320x320	W8A8	490.470	2.040	625.880	5.300

RV1126B

Model Name	Input	Quantization Method	Single Core FPS	Single Core Latency(ms)	Memory(MB)
yolov5m	640x640	W8A8	20.490	48.810	-
yolov5s	640x640	W8A8	47.690	20.970	-
yolov6n	640x640	W8A8	97.690	10.240	-
yolov6s	640x640	W8A8	30.870	32.400	-
yolov8s	640x640	W8A8	34.170	29.270	-
yolov9s	512x512	W8A8	42.600	23.470	-
yolox_s	640x384	W8A8	18.210	54.930	-
yolov9s	640x640	W8A8	30.150	33.170	-
ppyoloe_s	640x640	W8A8	39.890	25.070	-
retinaface-mobile320	320x320	W8A8	463.140	2.160	-
yolo11m	640x640	W8A8	12.450	80.300	-
yolo11n	640x640	W8A8	68.570	14.580	-
yolo11s	640x640	W8A8	32.260	30.990	-
yolov10n	640x640	W8A8	70.040	14.280	-
yolov10s	640x640	W8A8	33.750	29.630	-
yolov5n	640x640	W8A8	101.880	9.820	-
yolov5s_relu	640x640	W8A8	57.990	17.240	14.430
yolov6m	640x640	W8A8	15.390	64.980	46.390
yolov7	640x640	W8A8	11.180	89.470	55.530
yolov7-tiny	640x640	W8A8	66.290	15.090	11.980
yolov8m	640x640	W8A8	14.010	71.400	36.370
yolov8n	640x640	W8A8	78.470	12.740	7.520
yolov8n-obb	640x640	W8A8	79.040	12.650	7.470
yolox_s	640x640	W8A8	36.170	27.650	16.280

RK3566/RK3568

Model Name	Input	Quantization Method	Single Core FPS	Single Core Latency(ms)
yolov5s_relu	640x640	W8A8	25.500	39.220
yolov5n	640x640	W8A8	39.700	25.190
yolov5s	640x640	W8A8	19.300	51.810
yolov5m	640x640	W8A8	8.600	116.280
yolov6n	640x640	W8A8	48.800	20.490
yolov6s	640x640	W8A8	15.200	65.790
yolov6m	640x640	W8A8	7.200	138.890
yolov7-tiny	640x640	W8A8	27.900	35.840
yolov7	640x640	W8A8	4.600	217.390
yolov8n	640x640	W8A8	34.000	29.410
yolov8s	640x640	W8A8	15.100	66.230
yolov8m	640x640	W8A8	6.500	153.850
yolov8n-obb	640x640	W8A8	33.900	29.500
yolov10n	640x640	W8A8	20.700	48.310
yolov10s	640x640	W8A8	10.300	97.090
yolo11n	640x640	W8A8	20.600	48.540
yolo11s	640x640	W8A8	10.200	98.040
yolo11m	640x640	W8A8	4.600	217.390
yolox_s	640x640	W8A8	15.200	65.790
yolox_m	640x640	W8A8	6.600	151.520
ppyoloe_s	640x640	W8A8	7.400	135.140
ppyoloe_m	640x640	W8A8	4.100	243.900
yolov8n-pose	640x640	W8A8	22.600	44.250
retinaface-mobile320	320x320	W8A8	156.400	6.390
retinaface-resnet50-320	320x320	W8A8	18.700	53.480

RK3562

Model Name	Input	Quantization Method	Single Core FPS	Single Core Latency(ms)
yolov5s_relu	640x640	W8A8	33.200	30.120
yolov5n	640x640	W8A8	47.400	21.100
yolov5s	640x640	W8A8	23.600	42.370
yolov5m	640x640	W8A8	10.800	92.590
yolov6n	640x640	W8A8	56.400	17.730
yolov6s	640x640	W8A8	17.300	57.800
yolov6m	640x640	W8A8	8.600	116.280
yolov7-tiny	640x640	W8A8	36.500	27.400
yolov7	640x640	W8A8	5.900	169.490
yolov8n	640x640	W8A8	40.900	24.450
yolov8s	640x640	W8A8	18.400	54.350
yolov8m	640x640	W8A8	8.200	121.950
yolov8n-obb	640x640	W8A8	41.300	24.210
yolov10n	640x640	W8A8	34.100	29.330
yolov10s	640x640	W8A8	16.900	59.170
yolo11n	640x640	W8A8	34.000	29.410
yolo11s	640x640	W8A8	16.700	59.880
yolo11m	640x640	W8A8	6.500	153.850
yolox_s	640x640	W8A8	18.300	54.640
yolox_m	640x640	W8A8	8.200	121.950
ppyoloe_s	640x640	W8A8	20.000	50.000
ppyoloe_m	640x640	W8A8	9.200	108.700
yolov8n-pose	640x640	W8A8	31.000	32.260
retinaface-mobile320	320x320	W8A8	300.800	3.320
retinaface-resnet50-320	320x320	W8A8	26.900	37.170

RV1109

Model Name	Input	Quantization Method	Single Core FPS	Single Core Latency(ms)
yolov5s_relu	640x640	W8A8	20.200	49.500
yolov5n	640x640	W8A8	36.300	27.550
yolov5s	640x640	W8A8	13.600	73.530
yolov5m	640x640	W8A8	5.800	172.410
yolov6n	640x640	W8A8	37.800	26.460
yolov6s	640x640	W8A8	10.800	92.590
yolov6m	640x640	W8A8	5.600	178.570
yolov7-tiny	640x640	W8A8	15.400	64.940
yolov7	640x640	W8A8	3.300	303.030
yolov8n	640x640	W8A8	24.000	41.670
yolov8s	640x640	W8A8	8.900	112.360
yolov8m	640x640	W8A8	3.900	256.410
yolov8n-obb	640x640	W8A8	25.100	39.840
yolo11n	640x640	W8A8	11.700	85.470
yolo11s	640x640	W8A8	5.000	200.000
yolo11m	640x640	W8A8	2.800	357.140
yolox_s	640x640	W8A8	10.600	94.340
yolox_m	640x640	W8A8	4.600	217.390
ppyoloe_s	640x640	W8A8	11.200	89.290
ppyoloe_m	640x640	W8A8	5.200	192.310
retinaface-mobile320	320x320	W8A8	144.800	6.910
retinaface-resnet50-320	320x320	W8A8	14.600	68.490

RV1126

Model Name	Input	Quantization Method	Single Core FPS	Single Core Latency(ms)
yolov5s_relu	640x640	W8A8	29.200	34.250
yolov5n	640x640	W8A8	53.200	18.800
yolov5s	640x640	W8A8	20.000	50.000
yolov5m	640x640	W8A8	8.500	117.650
yolov6n	640x640	W8A8	56.800	17.610
yolov6s	640x640	W8A8	16.300	61.350
yolov6m	640x640	W8A8	8.300	120.480
yolov7-tiny	640x640	W8A8	22.400	44.640
yolov7	640x640	W8A8	4.800	208.330
yolov8n	640x640	W8A8	35.400	28.250
yolov8s	640x640	W8A8	13.100	76.340
yolov8m	640x640	W8A8	5.800	172.410
yolov8n-obb	640x640	W8A8	37.300	26.810
yolo11n	640x640	W8A8	17.000	58.820
yolo11s	640x640	W8A8	7.300	136.990
yolo11m	640x640	W8A8	4.000	250.000
yolox_s	640x640	W8A8	15.700	63.690
yolox_m	640x640	W8A8	6.800	147.060
ppyoloe_s	640x640	W8A8	16.400	60.980
ppyoloe_m	640x640	W8A8	7.700	129.870
retinaface-mobile320	320x320	W8A8	212.500	4.710
retinaface-resnet50-320	320x320	W8A8	20.800	48.080

RK1808

Model Name	Input	Quantization Method	Single Core FPS	Single Core Latency(ms)
yolov5s_relu	640x640	W8A8	37.200	26.880
yolov5n	640x640	W8A8	61.200	16.340
yolov5s	640x640	W8A8	28.200	35.460
yolov5m	640x640	W8A8	13.300	75.190
yolov6n	640x640	W8A8	66.800	14.970
yolov6s	640x640	W8A8	24.100	41.490
yolov6m	640x640	W8A8	11.500	86.960
yolov7-tiny	640x640	W8A8	37.200	26.880
yolov7	640x640	W8A8	7.400	135.140
yolov8n	640x640	W8A8	42.300	23.640
yolov8s	640x640	W8A8	19.100	52.360
yolov8m	640x640	W8A8	9.100	109.890
yolov8n-obb	640x640	W8A8	42.800	23.360
yolo11n	640x640	W8A8	17.600	56.820
yolo11s	640x640	W8A8	8.400	119.050
yolo11m	640x640	W8A8	5.100	196.080
yolox_s	640x640	W8A8	23.000	43.480
yolox_m	640x640	W8A8	10.700	93.460
ppyoloe_s	640x640	W8A8	21.100	47.390
ppyoloe_m	640x640	W8A8	9.400	106.380
retinaface-mobile320	320x320	W8A8	198.500	5.040
retinaface-resnet50-320	320x320	W8A8	24.600	40.650

4.2 Classification/Recognition Models

Below are the benchmarks for image classification, recognition, and OCR models on Rockchip NPU.

RK1820/RK1828

Model Name	Input	Quantization Method	Single Core FPS	Single-Core Latency(ms)	Multi-Core FPS	Memory(MB)
mobilenetv2	224x224	W8A8	280.450	3.566	1278.850	8.854
mobilenetv1	224x224	W8A8	377.060	2.652	1468.110	5.822
efficientnet_v2_m	240x240	W8A8	34.380	29.090	270.300	61.640
pfld	112x112	W8A8	409.150	2.444	1187.400	2.671
mobilenetv2	112x112	W8A8	648.920	1.541	1932.920	4.870
mobilenetv2	128x256	W8A8	374.560	2.670	1667.110	7.011
resnet34	128x256	W8A8	204.920	4.880	1358.490	24.404
iresnet50	112x112	W8A8	320.720	3.118	1759.990	26.063
resnet50-v2-7	224x224	W8A8	110.560	9.045	832.350	30.095

RK3572

Model Name	Input	Quantization Method	Single Core FPS	Single-Core Latency(ms)	Memory(MB)
mobilenetv2	224x224	W8A8	317.140	3.153	8.854
resnet50-v2-7	224x224	W8A8	109.030	9.172	30.095
yolo_world_v2s	640x640	W8A8	21.060	47.470	27.609

RK3588

Model Name	Input	Quantization Method	Single Core FPS	Single-Core Latency(ms)	Multi-Core FPS	Memory(MB)
mobilenetv1	224x224	W8A8	396.880	2.520	-	5.822
resnet50-v2-7	224x224	W8A8	111.680	8.954	291.120	30.095
mobilenetv2	224x224	W8A8	431.610	2.320	1233.370	14.570
lprnet	24x94	W8A8	586.400	1.710	-	-
ppocrv4_det	480x480	W8A8	50.700	19.720	-	-
ppocrv4_rec	48x320	FP16	73.900	13.530	-	-
lite-transformer-encoder-16	embedding-256_token-16	FP16	867.600	1.150	-	-
lite-transformer-decoder-16	embedding-256_token-16	FP16	343.800	2.910	-	-
yolo_world_v2s	640x640	W8A8	17.820	56.130	50.590	64.890

RK3576

Model Name	Input	Quantization Method	Single Core FPS	Single-Core Latency(ms)	Multi-Core FPS	Memory(MB)
yolo_world_v2s	640x640	FP16	14.610	68.460	26.490	96.680
mobilenetv2	224x224	W8A8	451.410	2.220	851.770	10.420
resnet50-v2-7	224x224	W8A8	95.290	10.490	176.550	53.760
lprnet	24x94	FP16	647.800	1.540	-	-
ppocrv4_det	480x480	W8A8	64.300	15.550	-	-
ppocrv4_rec	48x320	FP16	96.800	10.330	-	-
lite-transformer-encoder-16	embedding-256_token-16	FP16	784.100	1.280	-	-
lite-transformer-decoder-16	embedding-256_token-16	FP16	272.300	3.670	-	-
yolo_world_v2s	640x640	W8A8	23.780	42.050	41.600	52.070

RV1126B

Model Name	Input	Quantization Method	Single Core FPS	Single-Core Latency(ms)	Memory(MB)
mobilenetv2	224x224	W8A8	394.000	2.540	-
resnet50-v2-7	224x224	W8A8	83.050	12.040	-
yolo_world_v2s	640x640	W8A8	18.562	53.870	41.126

RK3566/RK3568

Model Name	Input	Quantization Method	Single Core FPS	Single-Core Latency(ms)
mobilenetv2	224x224	W8A8	180.700	5.530
resnet50-v2-7	224x224	W8A8	37.900	26.390
yolo_world_v2s	640x640	W8A8	7.400	135.140
lprnet	24x94	FP16	143.200	6.980
ppocrv4_det	480x480	W8A8	22.100	45.250
ppocrv4_rec	48x320	FP16	19.500	51.280
lite-transformer-encoder-16	embedding-256_token-16	FP16	337.500	2.960
lite-transformer-decoder-16	embedding-256_token-16	FP16	142.500	7.020

RK3562

Model Name	Input	Quantization Method	Single Core FPS	Single-Core Latency(ms)
mobilenetv2	224x224	W8A8	281.300	3.560
resnet50-v2-7	224x224	W8A8	54.900	18.210
yolo_world_v2s	640x640	W8A8	9.600	104.170
lprnet	24x94	FP16	420.600	2.380
ppocrv4_det	480x480	W8A8	28.000	35.710
ppocrv4_rec	48x320	FP16	54.300	18.420
lite-transformer-encoder-16	embedding-256_token-16	FP16	725.800	1.380
lite-transformer-decoder-16	embedding-256_token-16	FP16	252.000	3.970

RV1109

Model Name	Input	Quantization Method	Single Core FPS	Single-Core Latency(ms)
mobilenetv2	224x224	W8A8	212.900	4.700
resnet50-v2-7	224x224	W8A8	24.400	40.980
lprnet	24x94	W8A8	30.600	32.680
ppocrv4_det	480x480	W8A8	11.000	90.910
ppocrv4_rec	48x320	FP16	1.000	1000.000
lite-transformer-encoder-16	embedding-256_token-16	FP16	22.700	44.050
lite-transformer-decoder-16	embedding-256_token-16	FP16	48.000	20.830

RV1126

Model Name	Input	Quantization Method	Single Core FPS	Single-Core Latency(ms)
mobilenetv2	224x224	W8A8	322.300	3.100
resnet50-v2-7	224x224	W8A8	36.200	27.620
lprnet	24x94	W8A8	47.600	21.010
ppocrv4_det	480x480	W8A8	16.100	62.110
ppocrv4_rec	48x320	FP16	1.600	625.000
lite-transformer-encoder-16	embedding-256_token-16	FP16	35.400	28.250
lite-transformer-decoder-16	embedding-256_token-16	FP16	65.800	15.200

RK1808

Model Name	Input	Quantization Method	Single Core FPS	Single-Core Latency(ms)
mobilenetv2	224x224	W8A8	170.300	5.870
resnet50-v2-7	224x224	W8A8	37.100	26.950
lprnet	24x94	W8A8	30.100	33.220
ppocrv4_det	480x480	W8A8	14.200	70.420
ppocrv4_rec	48x320	FP16	6.700	149.250
lite-transformer-encoder-16	embedding-256_token-16	FP16	98.300	10.170
lite-transformer-decoder-16	embedding-256_token-16	FP16	109.900	9.100

4.3 Image Segmentation Models

Below are the benchmarks for image segmentation models on Rockchip NPU.

RK1820/RK1828

Model Name	Input	Quantization Method	Single Core FPS	Single Core Latency(ms)	Multi Core FPS	Memory(MB)
yolov8s-seg	640x640	W8A8	20.870	47.925	79.810	20.556

RK3572

Model Name	Input	Quantization Method	Single Core FPS	Single Core Latency(ms)	Memory(MB)
yolov8s-seg	640x640	W8A8	31.720	31.530	20.556
yolov5n-seg	640x640	W8A8	72.020	13.890	10.340
yolov5s-seg	640x640	W8A8	40.610	24.620	22.500
yolov5m-seg	640x640	W8A8	18.160	55.080	41.350
yolov8n-seg	640x640	W8A8	61.670	16.220	10.430
yolov8m-seg	640x640	W8A8	13.040	76.700	45.800
pp_liteseg_cityscapes	512x512	W8A8	25.490	39.230	36.930
mobilesam_encoder	448x448	FP16	16.170	61.840	42.470
mobilesam_decoder	112x112	FP16	133.270	7.500	11.110

RK3588

Model Name	Input	Quantization Method	Single Core FPS	Single Core Latency(ms)	Multi Core FPS	Memory(MB)
pp_liteseg_cityscapes	512x1024	W8A8	34.310	29.150	96.370	58.980
mobilesam_encoder	448x448	FP16	9.710	102.980	22.640	197.770
mobilesam_decoder	112x112	FP16	115.730	8.640	274.750	37.730
yolov5m-seg	640x640	W8A8	16.760	59.650	46.820	97.430
yolov5n-seg	640x640	W8A8	70.440	14.200	197.380	22.880
yolov5s-seg	640x640	W8A8	37.720	26.510	104.890	46.130
deeplab-v3-plus-mobilenet-v2	512x512	W8A8	34.000	29.410	-	-
ppseg_lite	512x512	W8A8	35.700	28.010	-	-
mobilesam_encoder_tiny	640x640	FP16	10.000	100.000	-	-
yolov8m-seg	640x640	W8A8	12.270	81.530	35.610	112.730
yolov8n-seg	640x640	W8A8	59.820	16.720	168.850	22.740
yolov8s-seg	640x640	W8A8	28.837	34.680	82.660	55.160

RK3576

Model Name	Input	Quantization Method	Single Core FPS	Single Core Latency(ms)	Multi Core FPS	Memory(MB)
pp_liteseg_cityscapes	512x1024	W8A8	48.710	20.530	76.500	35.380
mobilesam_encoder	448x448	FP16	10.680	93.640	17.320	169.720
deeplab-v3-plus-mobilenet-v2	513x513	W8A8	39.400	25.380	-	-
ppseg_lite	512x512	W8A8	33.600	29.760	-	-
mobilesam_encoder_tiny	448x448	FP16	11.900	84.030	-	-
yolov5s-seg	640x640	W8A8	41.030	24.370	67.490	32.800
yolov5m-seg	640x640	W8A8	17.120	58.400	30.020	66.600
yolov5n-seg	640x640	W8A8	92.080	10.860	139.790	17.700
yolov8n-seg	640x640	W8A8	73.310	13.640	105.630	18.800
yolov8m-seg	640x640	W8A8	11.910	83.960	20.610	80.400
yolov8s-seg	640x640	W8A8	30.230	33.080	46.440	40.100
mobilesam_decoder	112x112	FP16	102.610	9.750	151.420	33.210

RV1126B

Model Name	Input	Quantization Method	Single Core FPS	Single Core Latency(ms)	Memory(MB)
yolov8s-seg	640x640	W8A8	25.990	38.470	-
yolov5n-seg	640x640	W8A8	80.390	12.440	14.230
yolov5s-seg	640x640	W8A8	36.570	27.340	29.720
yolov5m-seg	640x640	W8A8	15.510	64.500	63.410
yolov8n-seg	640x640	W8A8	62.500	16.000	14.930
yolov8m-seg	640x640	W8A8	10.920	91.590	74.350
pp_liteseg_cityscapes	512x512	W8A8	12.810	78.070	35.870
mobilesam_encoder	448x448	FP16	9.480	105.470	104.480
mobilesam_decoder	112x112	FP16	84.580	11.820	27.790
pp_liteseg_cityscapes	512x1024	W8A8	34.290	29.160	34.162

RK3566/RK3568

Model Name	Input	Quantization Method	Single Core FPS	Single Core Latency(ms)
deeplab-v3-plus-mobilenet-v2	513x513	W8A8	10.900	91.740
yolov5n-seg	640x640	W8A8	32.200	31.060
yolov5s-seg	640x640	W8A8	15.000	66.670
yolov5m-seg	640x640	W8A8	6.800	147.060
yolov8n-seg	640x640	W8A8	27.800	35.970
yolov8s-seg	640x640	W8A8	11.700	85.470
yolov8m-seg	640x640	W8A8	5.200	192.310
ppseg_lite	512x512	W8A8	5.900	169.490
mobilesam_encoder_tiny	448x448	FP16	1.000	1000.000
mobilesam_decoder	112x112	FP16	24.300	41.150

RK3562

Model Name	Input	Quantization Method	Single Core FPS	Single Core Latency(ms)
deeplab-v3-plus-mobilenet-v2	513x513	W8A8	21.400	46.730
ppseg_lite	512x512	W8A8	13.900	71.940
mobilesam_encoder_tiny	448x448	FP16	6.600	151.520
yolov5n-seg	640x640	W8A8	40.130	24.920
yolov5s-seg	640x640	W8A8	18.040	55.430
yolov5m-seg	640x640	W8A8	8.300	120.550
yolov8n-seg	640x640	W8A8	33.350	29.980
yolov8s-seg	640x640	W8A8	13.770	72.600
yolov8m-seg	640x640	W8A8	6.200	161.250
pp_liteseg_cityscapes	512x512	W8A8	10.640	93.960

Model Name	Input	Quantization Method	Single Core FPS	Single Core Latency(ms)
pp_liteseg_cityscapes	512x512	W8A8	10.040	99.900
mobilesam_encoder	448x448	FP16	6.580	152.030
mobilesam_decoder	112x112	FP16	71.120	14.060

RV1109

Model Name	Input	Quantization Method	Single Core FPS	Single Core Latency(ms)
deeplab-v3-plus-mobilenet-v2	513x513	W8A8	10.100	99.010
yolov5n-seg	640x640	W8A8	28.600	34.970
yolov5s-seg	640x640	W8A8	9.600	104.170
yolov5m-seg	640x640	W8A8	4.700	212.770
yolov8n-seg	640x640	W8A8	18.600	53.760
yolov8s-seg	640x640	W8A8	6.600	151.520
yolov8m-seg	640x640	W8A8	3.100	322.580
ppseg_lite	512x512	W8A8	18.400	54.350

RV1126

Model Name	Input	Quantization Method	Single Core FPS	Single Core Latency(ms)
deeplab-v3-plus-mobilenet-v2	513x513	W8A8	13.000	76.920
yolov5n-seg	640x640	W8A8	42.200	23.700
yolov5s-seg	640x640	W8A8	14.000	71.430
yolov5m-seg	640x640	W8A8	6.800	147.060
yolov8n-seg	640x640	W8A8	27.600	36.230
yolov8s-seg	640x640	W8A8	9.800	102.040
yolov8m-seg	640x640	W8A8	4.600	217.390
ppseg_lite	512x512	W8A8	27.100	36.900

RK1808

Model Name	Input	Quantization Method	Single Core FPS	Single Core Latency(ms)
deeplab-v3-plus-mobilenet-v2	513x513	W8A8	4.400	227.270
yolov5n-seg	640x640	W8A8	49.600	20.160
yolov5s-seg	640x640	W8A8	22.500	44.440
yolov5m-seg	640x640	W8A8	10.800	92.590
yolov8n-seg	640x640	W8A8	32.900	30.400
yolov8s-seg	640x640	W8A8	14.600	68.490
yolov8m-seg	640x640	W8A8	6.900	144.930
ppseg_lite	512x512	W8A8	20.900	47.850

4.4 ViT/CLIP Models

Below are the benchmarks for Vision Transformer, CLIP, and other models on Rockchip NPU.

RK1820/RK1828

Model Name	Input	Quantization Method	Single Core FPS	Single-Core Latency(ms)	Multi-Core FPS	Memory(MB)
vit-base-patch14	224x224	FP16	13.850	72.219	179.770	171.964
vit-base-patch14	224x224	W4A16	23.110	43.279	180.420	49.790
vit-base-patch16	224x224	FP16	19.090	52.391	151.160	169.625
vit-base-patch16	224x224	W4A16	31.920	31.331	251.430	47.450
vit-large-patch14	224x224	W4A16	7.710	129.754	61.460	156.914
vit-large-patch16	224x224	W4A16	10.600	94.357	84.290	153.794
dinov3_vits16	224x224	FP16	44.740	22.350	254.350	45.540
dinov3_vitb16	224x224	FP16	16.920	59.100	119.180	168.890
dinov2_vits14	224x224	FP16	37.540	26.641	264.590	45.709
dinov2_vitb14	224x224	FP16	12.810	78.038	99.370	171.719
dinov3_vits16	624x624	FP16	2.420	414.070	18.930	183.055
vit-large-patch16	224x224	FP16	26.590	37.612	47.340	611.839
siglip-so400m-patch14-384	384x384	FP16	4.980	200.853	-	918.091
siglip2-so400m-patch14-384	384x384	FP16	4.980	200.723	-	918.091
vit-large-patch14	224x224	FP16	4.660	214.760	36.640	589.310

RK3572

Model Name	Input	Quantization Method	Single Core FPS	Single-Core Latency(ms)	Memory(MB)
cn-clip-large-patch14-vision	224x224	W4A16	5.930	168.680	179.352
cn-clip-large-patch14-vision	224x224	FP16	3.940	253.950	589.918
cn-clip-large-patch14-vision	336x336	W4A16	2.060	484.380	208.063
cn-clip-large-patch14-vision	336x336	FP16	1.510	663.140	618.629
siglip-so400m-patch14-384	384x384	W4A16	0.920	1084.080	320.062
siglip-so400m-patch14-384	384x384	FP16	0.700	1426.030	876.320
siglip2-base-patch16-512	512x512	W4A16	2.610	382.750	133.366
siglip2-base-patch16-512	512x512	FP16	2.040	491.230	257.336
siglip2-so400m-patch14-384	384x384	W4A16	0.920	1082.780	320.062
siglip2-so400m-patch14-384	384x384	FP16	0.710	1415.160	876.320
vit-base-patch14	224x224	W4A16	17.910	55.830	56.135
vit-base-patch14	224x224	FP16	11.600	86.180	172.338
vit-base-patch16	224x224	W4A16	25.920	38.580	53.708
vit-base-patch16	224x224	FP16	16.470	60.730	169.911

Model Name	Input	Quantization Method	Single Core FPS	Single-Core Latency(ms)	Memory(MB)
vit-large-patch14	224x224	W4A16	6.140	162.950	178.931
vit-large-patch14	224x224	FP16	4.020	248.550	589.806
vit-large-patch16	224x224	W4A16	8.360	119.660	175.694
vit-large-patch16	224x224	FP16	5.160	193.670	586.569
dinov2_vits14	224x224	W4A16	42.070	23.770	17.078
dinov2_vits14	224x224	FP16	33.850	29.540	45.859
dinov2_vitb14	224x224	W4A16	17.020	58.760	56.167
dinov2_vitb14	224x224	FP16	11.290	88.570	171.339
dinov3_vits16	224x224	W4A16	47.940	20.860	16.902
dinov3_vits16	224x224	FP16	39.340	25.420	45.684
dinov3_vitb16	224x224	W4A16	21.380	46.780	53.405
dinov3_vitb16	224x224	FP16	14.490	69.030	168.577
dinov3_vits16	640x640	W4A16	2.310	432.340	109.796
dinov3_vits16	640x640	FP16	2.160	463.490	138.577
EVA02-Large	224x224	W4A16	2.930	341.780	188.952
EVA02-Large	224x224	FP16	2.380	420.770	598.022
cn-clip-large-patch14-text	1x52	FP16	62.030	16.120	195.360

RK3588

Model Name	Input	Quantization Method	Single Core FPS	Single-Core Latency(ms)	Multi-Core FPS	Memory(MB)
fastvit_sa12	224x224	FP16	51.540	19.400	132.720	160.150
cn-clip-large-patch14-vision	224x224	FP16	3.420	292.150	6.250	1799.890
cn-clip-large-patch14-text	1x52	FP16	51.400	19.450	99.700	593.020
cn-clip-large-patch14-vision	336x336	FP16	1.090	921.770	1.870	1852.050
dinov2_vitb14	224x224	FP16	9.790	102.180	18.240	524.670
dinov2_vits14	224x224	FP16	21.750	45.990	43.890	142.630
dinov3_vits16	224x224	FP16	20.490	48.820	42.070	146.640
dinov3_vits16	640x640	FP16	1.120	891.740	2.290	236.290
dinov3_vitb16	224x224	FP16	10.270	97.340	19.440	528.720
EVA02-Large	224x224	FP16	2.830	353.320	5.240	1848.430
siglip2-base-patch16-512	512x512	FP16	1.170	857.750	2.170	646.100
siglip2-so400m-patch14-384	384x384	FP16	0.580	1712.300	0.990	2617.450
vit-base-patch14	224x224	FP16	9.960	100.450	18.710	530.060
vit-base-patch16	224x224	FP16	11.610	86.120	21.910	523.360
vit-large-patch16	224x224	FP16	4.460	224.210	8.240	1794.970
clip-vit-base-patch32-text	1x20	W8A8	95.800	10.440	-	-
clip-vit-base-patch32-vision	224x224	FP16	6.500	153.850	-	-
clip-vit-base-patch32-text	1x20	FP16	96.000	10.420	-	-
vit-large-patch14	224x224	FP16	1.590	628.860	4.070	1792.780
vit-large-patch14	224x224	W8A8	3.368	296.910	8.350	910.540

RK3576

Model Name	Input	Quantization Method	Single Core FPS	Single-Core Latency(ms)	Multi-Core FPS	Memory(MB)
fastvit_sa12	224x224	FP16	51.140	19.560	93.090	107.870
cn-clip-large-patch14-vision	224x224	FP16	29.330	34.090	29.880	405.800
cn-clip-large-patch14-vision	336x336	FP16	28.650	34.910	28.980	405.800
dinov2_vitb14	518x518	FP16	6.290	158.880	8.610	355.200
dinov2_vits14	518x518	FP16	19.880	50.310	23.480	99.500
dinov3_vits16	518x518	FP16	17.340	57.660	22.270	100.000
dinov3_vitb16	224x224	FP16	6.820	146.700	9.050	355.500
dinov3_vits16	640x640	FP16	17.050	58.660	22.250	100.000
EVA02-Large	224x224	FP16	2.170	460.970	2.870	1248.300
siglip-so400m-patch14-384	384x384	FP16	0.500	2014.080	0.700	1776.200
siglip2-base-patch16-512	512x512	FP16	0.940	1069.780	1.370	463.900
siglip2-so400m-patch14-384	384x384	FP16	0.500	2012.840	0.700	1776.200
vit-base-patch14	224x224	FP16	6.510	153.560	8.820	357.400
vit-base-patch16	224x224	FP16	7.380	135.450	10.070	355.500
vit-large-patch16	224x224	FP16	2.440	409.230	3.530	1220.500
cn-clip-large-patch14-vision	224x224	W4A16	3.630	275.620	4.400	647.900
cn-clip-large-patch14-vision	336x336	W4A16	1.250	799.760	1.690	688.600
dinov2_vitb14	518x518	W4A16	10.580	94.570	12.320	192.400
dinov2_vits14	518x518	W4A16	25.510	39.210	30.290	58.900
dinov3_vits16	518x518	W4A16	25.870	38.660	31.200	59.800
dinov3_vitb16	224x224	W4A16	11.410	87.620	14.400	193.800
dinov3_vits16	640x640	W4A16	25.780	38.790	30.950	59.800
EVA02-Large	224x224	W4A16	3.270	305.940	3.610	670.100
siglip-so400m-patch14-384	384x384	W4A16	0.550	1812.140	0.710	1233.400
siglip2-base-patch16-512	512x512	W4A16	1.270	784.890	1.700	279.700

Model Name	Input	Quantization Method	Single Core FPS	Single-Core Latency(ms)	Multi-Core FPS	Memory(MB)
siglip2-so400m-patch14-384	384x384	W4A16	0.550	1811.300	0.710	1233.500
vit-base-patch14	224x224	W4A16	11.080	90.230	12.790	193.200
vit-base-patch16	224x224	W4A16	12.410	80.610	16.350	191.000
clip-vit-base-patch32-text	1x20	FP16	64.850	15.420	112.070	254.150
clip-vit-base-patch32-vision	224x224	FP16	10.041	99.590	10.290	374.835
vit-large-patch14	224x224	FP16	2.954	338.490	3.030	1223.832
vit-large-patch14	224x224	W4A16	4.395	227.530	4.520	647.472
vit-large-patch16	224x224	W4A16	5.666	176.500	5.860	643.046

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Model Name	Input	Quantization Method	Single Core FPS	Single-Core Latency(ms)	Memory(MB)
cn-clip-large-patch14-text	1x52	FP16	28.310	35.320	396.410
cn-clip-large-patch14-text	1x52	W4A16	40.000	25.000	152.720
cn-clip-large-patch14-vision	224x224	FP16	24.780	40.360	195.750
cn-clip-large-patch14-vision	336x336	FP16	24.770	40.380	195.750
dinov2_vitb14	518x518	FP16	5.410	184.930	169.320
dinov2_vits14	518x518	FP16	16.410	60.930	45.120
dinov3_vits16	640x640	FP16	0.880	1136.560	132.680
dinov3_vitb16	224x224	FP16	5.910	169.080	169.010
dinov3_vits16	224x224	FP16	14.420	69.360	45.440
EVA02-Large	196x196	FP16	1.750	572.860	592.520

Model Name	Input	Quantization Method	Single Core FPS	Single-Core Latency(ms)	Memory(MB)
siglip2-base-patch16-512	512x512	FP16	0.810	1238.740	218.250
siglip2-so400m-patch14-384	384x384	FP16	0.310	3189.080	850.610
siglip-so400m-patch14-384	384x384	FP16	0.310	3204.850	850.610
vit-base-patch14	224x224	FP16	5.600	178.660	169.930
vit-base-patch16	224x224	FP16	6.590	151.640	170.530
vit-large-patch14	224x224	FP16	1.750	572.880	586.920
vit-large-patch16	224x224	FP16	2.140	467.880	587.850
cn-clip-large-patch14-vision	224x224	W4A16	3.040	328.560	295.930
cn-clip-large-patch14-vision	336x336	W4A16	0.990	1015.050	305.290
dinov2_vitb14	518x518	W4A16	9.000	111.090	86.340
dinov2_vits14	518x518	W4A16	21.440	46.640	24.160
dinov3_vits16	640x640	W4A16	0.990	1011.240	97.020
dinov3_vitb16	224x224	W4A16	9.930	100.730	85.690
dinov3_vits16	224x224	W4A16	21.130	47.330	23.840
EVA02-Large	196x196	W4A16	2.610	382.820	299.830
siglip2-base-patch16-512	512x512	W4A16	1.050	951.340	122.010
siglip-so400m-patch14-384	384x384	W4A16	0.490	2063.730	434.560
vit-base-patch14	224x224	W4A16	9.280	107.760	86.590
vit-base-patch16	224x224	W4A16	11.170	89.540	86.050
vit-large-patch14	224x224	W4A16	3.090	323.450	295.870
vit-large-patch16	224x224	W4A16	4.130	242.010	295.790
fastvit_sa12	224x224	FP16	51.351	19.470	101.855
clip-vit-base-patch32-text	1x20	FP16	60.813	16.440	246.862

RK3566/RK3568

Model Name	Input	Quantization Method	Single Core FPS	Single-Core Latency(ms)
clip-vit-base-patch32-text	1x20	FP16	29.800	33.560
clip-vit-base-patch32-vision	224x224	FP16	2.300	434.780

RK3562

Model Name	Input	Quantization Method	Single Core FPS	Single-Core Latency(ms)
clip-vit-base-patch32-text	1x20	FP16	67.400	14.840
clip-vit-base-patch32-vision	224x224	FP16	3.400	294.120

5. LLM Model Performance

Below are the benchmarks for large language models on Rockchip NPU.

> Metrics: TTFT = Time to First Token, Prefill = Prefill Speed, TPS = Decoding Speed (tokens/s), TPOT = Time Per Output Token

RK1820/RK1828

Model Name	Quantization Method	Context	seq_len	new_tokens	TTFT(ms)	Prefill(tok/s)	TPOT(ms)	TPS(tok/s)	Memory(MB)
Llama-3.2-3B-Instruct	W4A16	1024	128	128	86.190	1485.087	9.130	109.528	2100.000
Qwen2.5-0.5B-Instruct	W4A16	4096	4096	128	1090.772	3755.140	7.084	141.172	575.000
Qwen2.5-1.5B-Instruct	W4A16	4096	1024	128	439.967	2327.445	8.345	119.832	1485.000
Qwen2.5-1.5B-Instruct	W4A16	4096	4096	128	2134.989	1918.511	11.017	90.770	1485.000
Qwen2.5-3B-Instruct	W4A16	4096	1024	128	740.977	1381.959	11.646	85.864	2518.000
Qwen2.5-3B-Instruct	W4A16	4096	4096	128	3486.575	1174.792	15.171	65.917	2518.000
Qwen3-0.6B	W4A16	4096	1024	128	271.747	3768.214	7.241	138.098	905.000
Qwen3-0.6B	W4A16	4096	4096	128	1488.687	2751.418	9.931	100.696	905.000
Qwen3-1.7B	W4A16	4096	1024	128	459.018	2230.847	8.950	111.731	1568.000
Qwen3-1.7B	W4A16	4096	4096	128	2337.350	1752.412	11.935	83.789	1568.000
Qwen2.5-0.5B-Instruct	W4A16	4096	1024	1024	206.932	4948.489	5.635	177.475	575.000
Qwen2.5-1.5B-Instruct	W4A16	4096	1024	1024	446.971	2290.976	8.482	117.903	1485.000
Qwen2.5-3B-Instruct	W4A16	4096	1024	1024	744.875	1374.728	11.763	85.016	2518.000
Qwen3-0.6B	W4A16	4096	1024	1024	273.595	3742.762	7.398	135.171	905.000
Qwen3.5-2B	W4A16	16384	128	128	364.699	350.974	24.370	41.035	2408.000
Qwen2.5-0.5B-Instruct	W4A16	4096	128	128	22.281	5744.714	4.731	211.383	575.000
Qwen3-0.6B	W4A16	4096	128	128	28.270	4527.823	5.743	174.116	905.000
Qwen3-1.7B	W4A16	4096	128	128	53.446	2394.935	7.416	134.841	1568.000
Qwen2.5-3B-Instruct	W4A16	4096	128	128	84.800	1509.430	9.772	102.336	2518.000
Qwen2.5-3B	W4A16	1024	128	128	85.428	1498.340	9.953	100.468	2220.000
Qwen3-4B	W4A16	4096	128	128	108.186	1183.153	11.546	86.611	2994.000
Qwen2.5-7B	W4A16	1024	128	128	160.246	798.771	14.472	69.099	4400.000
Qwen3-4B-Instruct-2507	W4A16	38912	128	128	114.139	1121.444	12.537	79.763	4837.000
Qwen3-8B	W4A16	1024	128	128	180.079	710.799	16.576	60.330	4712.000
Qwen2.5-7B-Instruct	W4A16	4096	128	128	159.687	801.569	14.293	69.966	4638.000
Qwen2.5-1.5B-Instruct	W4A16	4096	128	128	47.443	2697.993	6.805	146.954	1485.000
Qwen2.5-7B-Instruct	W4A16	1024	128	128	159.968	800.160	14.597	68.509	4441.680
Qwen3-4B-Instruct-2507	W4A16	1024	128	128	106.619	1200.538	11.687	85.569	2586.000
Qwen3-4B	W4A16	1024	128	128	106.435	1202.607	11.534	86.699	-
Qwen3-4B-Instruct-2507	W4A16	16384	8192	128	13181.810	621.462	16.557	60.391	-
Qwen2.5-7B-Instruct	W4A16	4096	1024	128	1385.620	739.019	16.059	62.271	4638.000
Qwen2.5-7B-Instruct	W4A16	4096	4096	128	6339.583	646.099	18.854	53.040	4638.000
Qwen3-4B	W4A16	4096	1024	128	979.224	1045.726	13.548	73.814	2994.000
Qwen3-4B	W4A16	4096	4096	128	5055.826	810.154	17.023	58.744	2994.000
Qwen2.5-7B-Instruct	W4A16	4096	1024	1024	1387.454	738.042	15.984	62.563	4638.000
Qwen3-1.7B	W4A16	4096	1024	1024	484.357	2114.144	9.046	110.542	1568.000
Qwen3-4B	W4A16	4096	1024	1024	948.380	1079.736	13.587	73.598	2994.000

RK3572

Model Name	Quantization Method	Context	seq_len	new_tokens	TTFT(ms)	Prefill(tok/s)	TPOT(ms)	TPS(tok/s)	Memory(MB)
Qwen2.5-0.5B-Instruct	W4A16	16384	128	128	101.450	1261.750	21.760	45.960	304.000
Qwen3-0.6B	W4A16	16384	128	128	166.310	769.650	36.960	27.060	476.000
Qwen2.5-1.5B-Instruct	W4A16	16384	128	128	278.570	459.500	52.830	18.930	861.000
Qwen3-1.7B	W4A16	16384	128	128	300.850	425.460	69.870	14.310	1066.000

RK3588

Model Name	Quantization Method	Context	seq_len	new_tokens	TTFT(ms)	Prefill(tok/s)	TPOT(ms)	TPS(tok/s)	Memory(MB)
Qwen2.5-0.5B	W8A8	1024	128	128	163.690	781.990	26.030	38.410	657.140
Qwen3-0.6B	W8A8	1024	128	128	372.400	343.720	35.510	28.160	855.710
Qwen2.5-1.5B-Instruct	W8A8	1024	128	128	409.580	312.520	62.510	16.000	1688.230
Qwen3-1.7B	W8A8	1024	128	128	471.690	271.370	72.680	13.760	1934.470
Qwen3.5-2B	W8A8	1024	128	128	961.070	133.180	81.160	12.330	2109.400
Qwen3-4B-Instruct-2507	W8A8	1024	128	128	1080.800	118.430	151.200	6.610	4183.900

RK3576

Model Name	Quantization Method	Context	seq_len	new_tokens	TTFT(ms)	Prefill(tok/s)	TPOT(ms)	TPS(tok/s)	Memory(MB)
Qwen2.5-0.5B	W4A16	1024	128	128	368.460	347.390	34.720	28.800	492.680
Qwen3-0.6B	W4A16	1024	128	128	624.900	204.830	49.070	20.380	418.860
Qwen2.5-1.5B-Instruct	W4A16	1024	128	128	907.710	141.010	81.040	12.340	565.830
Qwen3-1.7B	W4A16	1024	128	128	788.210	162.390	91.790	10.890	1271.270
Qwen3-1.7B	W8A8	1024	128	128	811.640	157.710	135.960	7.360	1940.300
Qwen3.5-2B	W4A16	1024	128	128	1692.920	75.610	109.120	9.160	1462.740
Qwen3.5-2B	W8A8	1024	128	128	1703.610	75.130	151.880	6.580	2113.870
Qwen3-4B-Instruct-2507	W4A16	1024	128	128	1758.370	72.790	185.580	5.390	2460.340
Qwen3-4B-Instruct-2507	W8A8	1024	128	128	1673.120	76.500	321.620	3.110	4292.100

RV1126B

Model Name	Quantization Method	Context	seq_len	new_tokens	TTFT(ms)	Prefill(tok/s)	TPOT(ms)	TPS(tok/s)	Memory(MB)
Qwen2.5-0.5B	W4A16	1024	128	128	528.850	242.030	46.230	21.630	650.830
Qwen3-0.6B	W4A16	1024	128	128	814.770	157.100	78.270	12.780	1071.120
Qwen2.5-1.5B-Instruct	W4A16	1024	128	128	1440.120	88.880	120.750	8.280	946.860
Qwen3-1.7B	W4A16	1024	128	128	1564.230	81.830	137.360	7.280	1108.100

6. VLM Model Performance

The following are the benchmark for the visual language models on the Rockchip NPU.

> Metrics: Vision Latency = Visual encoding latency, TTFT = Time to First Token, TPS = Decoding Speed (tokens/s)

RK1820/RK1828

Model	Quantization Method	Input	Vision Time(ms)	TTFT(ms)	TPS(tok/s)	Memory(MB)
Qwen3-VL-4B	W4A16	[1,3,640,640]	548.000	585.000	61.000	1760.000
Qwen3-VL-4B	W4A16	[1,3,384,384]	160.000	142.827	87.295	2722.600
Qwen2.5-Omni-3B	W4A16	[1,3,392,392]	318.000	91.900	104.000	2640.000
InternVL3-2B	W4A16	[1,3,448,448]	178.000	201.560	145.235	1407.101
Qwen3-VL-2B	W4A16	[1,3,384,384]	154.000	185.063	140.623	1408.150
MiniCPM-V-4	W4A16	[1,3,448,448]	224.000	97.225	110.875	2612.394
Qwen2.5-VL-7B_prune	W4A16	[1,3,392,392]	272.000	171.465	69.194	-
InternVL3.5-4B	W4A16	[1,3,448,448]	181.000	441.343	87.862	2823.019
Qwen2.5-VL-3B	W4A16	[1,3,384,384]	272.000	259.789	130.836	2541.502
FastVLM-1.5B	W4A16	[1,3,512,512]	127.000	97.555	135.613	1328.797
Qwen2.5-VL-3B	W4A16	[1,3,392,392]	269.000	261.405	126.517	2541.502

RK3572

Model	Quantization Method	Input	Vision Time(ms)	TTFT(ms)	TPS(tok/s)	Memory(MB)
Qwen3-VL-2B-Instruct	W4A16	[1,3,384,384]	536.000	328.700	13.000	1421.000